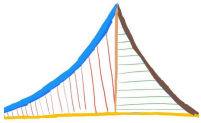
Math for EE



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. 10

A. Formulae

1.1.1 Tangent–radius property. If PA is a tangent to a circle with centre O at the point A,

$$OA \perp PA \tag{1.1.1.1}$$

1.1.2 Tangents from an external point. If PA and PB are tangents from P to a circle with centre O,

$$PA = PB, \quad \triangle OAP \cong \triangle OBP, \quad \angle POA = \angle POB \tag{1.1.2.1}$$

Solution: $OA = OB$ (radii), $PA = PB$ and OP is common, so the triangles are congruent by SSS.

1.1.3 Angle sum of a quadrilateral and of a triangle,

$$\sum \text{angles} = 360^\circ \text{ (quadrilateral)}, \quad 180^\circ \text{ (triangle)} \tag{1.1.3.1}$$

1.1.4 Angle in a semicircle. If AB is a diameter and C lies on the circle,

$$\angle ACB = 90^\circ \tag{1.1.4.1}$$

1.1.5 Tangent–chord theorem. The angle between a tangent and a chord through the point of contact equals the angle in the alternate segment. If PC is tangent at C,

$$\angle PCB = \angle CAB \tag{1.1.5.1}$$

1.1.6 Section formula. The point $\mathbf{R}$ dividing $\mathbf{P}$ and $\mathbf{Q}$ internally in the ratio $m : n$,

$$\mathbf{R} = \frac{m\mathbf{Q} + n\mathbf{P}}{m+n} = \mathbf{P} + \frac{m}{m+n}(\mathbf{Q} - \mathbf{P}) \tag{1.1.6.1}$$

For $m = n$ this is the mid-point $\frac{\mathbf{P}+\mathbf{Q}}{2}$.

1.1.7 Squared length and perpendicularity,

$$\|\mathbf{A}\|^2 = \mathbf{A}^\top \mathbf{A}, \quad \mathbf{A} \perp \mathbf{B} \iff \mathbf{A}^\top \mathbf{B} = 0 \tag{1.1.7.1}$$

1.1.8 Angle between two vectors. For direction vectors $\mathbf{m}_1$ and $\mathbf{m}_2$,

$$\cos\theta = \frac{\mathbf{m}_1^\top \mathbf{m}_2}{\|\mathbf{m}_1\| \, \|\mathbf{m}_2\|} \tag{1.1.8.1}$$

1.1.9 Altitude to the hypotenuse. If $\angle BAC = 90^\circ$ and $AD \perp BC$, then

$$AD^2 = BD \cdot DC \tag{1.1.9.1}$$

Solution: Take $\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} BD \\ 0 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} BD + DC \\ 0 \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} BD \\ h \end{pmatrix}$. From (1.1.7.1),

$$(\mathbf{B} - \mathbf{A})^\top (\mathbf{C} - \mathbf{A}) = -BD \cdot DC + h^2 = 0 \tag{1.1.9.2}$$

1.1.10 Zeroes of a quadratic. For $ax^2 + bx + c$ with zeroes α, β,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a} \tag{1.1.10.1}$$

1.1.11 Newton–Raphson method. To solve $f(x) = 0$,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{1.1.11.1}$$

Solution: The tangent to f at x_n is $y = f(x_n) + f'(x_n)(x - x_n)$. Setting $y = 0$ gives x_{n+1}.

1.1.12 Arithmetic progression with first term a and common difference d,

$$a_n = a + (n-1)d, \quad S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + a_n) \tag{1.1.12.1}$$

1.1.13 Gaussian elimination and rank. For a system $\mathbf{Ax} = \mathbf{b}$ with n unknowns, row-reduce the augmented matrix $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$. The system is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix} \tag{1.1.13.1}$$

and the solution is unique when this common rank equals n.

1.1.14 Simple interest on a principal P at rate $r\%$ per annum for t years,

$$I = \frac{Prt}{100} \tag{1.1.14.1}$$

1.1.15 Standard values and inverse sine,

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{1}{\sqrt{3}}, \quad \tan 45^\circ = 1 \tag{1.1.15.1}$$

The principal value $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$.

1.1.16 Angle of depression. From $\mathbf{B}$ at height h above the origin, the line of sight to a ground point with angle of depression θ has direction $\mathbf{m} = \begin{pmatrix}\cos\theta \\ -\sin\theta\end{pmatrix}$ and meets the ground $\mathbf{e}_2^{\top}\mathbf{x} = 0$ at

$$\mathbf{B} + \lambda\mathbf{m}, \quad \lambda = -\frac{\mathbf{e}_2^{\top}\mathbf{B}}{\mathbf{e}_2^{\top}\mathbf{m}} = \frac{h}{\sin\theta} \tag{1.1.16.1}$$

1.1.17 Probability from a pmf. For a random variable X with pmf $p_X(x)$ and an event E,

$$\Pr(E) = \sum_{x \in E} p_X(x), \qquad \sum_x p_X(x) = 1 \tag{1.1.17.1}$$

1.1.18 Unit step, convolution and the z-transform (used for sequences such as an AP). The unit step is $u[n] = 1$ for $n \geq 0$ and 0 otherwise, and

$$(x * h)[n] = \sum_k x[k]\, h[n-k], \quad X(z) = \sum_n x[n]\, z^{-n} \tag{1.1.18.1}$$

$$(x * h)[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)\, H(z) \tag{1.1.18.2}$$

The running sum of a sequence is its convolution with $u[n]$. Useful transform pairs (for $n \geq 0$) are

$$u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-z^{-1}}, \quad n\,u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-1}}{\left(1-z^{-1}\right)^2} \tag{1.1.18.3}$$

$$(n+1)\,u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^2}, \quad \frac{n\,(n+1)}{2}u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-1}}{\left(1-z^{-1}\right)^3} \tag{1.1.18.4}$$

B. Problems

1.2.1 If tangents PA and PB drawn from an external point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is equal to (10, 2025)

a) 40° b) 50° c) 60° d) 80°

Solution: Take the centre $\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix}$ as the origin and $\mathbf{P} = \begin{pmatrix}d\\0\end{pmatrix}$ on the x-axis. Let r be the radius and $\phi = \angle POA$. The points of contact are symmetric about OP, so

$$\mathbf{A} = r\begin{pmatrix}\cos\phi\\ \sin\phi\end{pmatrix}, \quad \mathbf{B} = r\begin{pmatrix}\cos\phi\\ -\sin\phi\end{pmatrix} \tag{1.2.1.1}$$

The radius is perpendicular to the tangent at the point of contact,

$$(\mathbf{A}-\mathbf{O})^\top(\mathbf{A}-\mathbf{P}) = 0 \implies \|\mathbf{A}\|^2 = \mathbf{A}^\top\mathbf{P} \implies r^2 = rd\cos\phi \implies r = d\cos\phi \tag{1.2.1.2}$$

Hence the direction vectors of the tangents PA and PB are

$$\mathbf{A}-\mathbf{P} = d\begin{pmatrix}\cos^2\phi - 1\\ \sin\phi\cos\phi\end{pmatrix} = d\sin\phi\begin{pmatrix}-\sin\phi\\ \cos\phi\end{pmatrix} \tag{1.2.1.3}$$

$$\mathbf{B}-\mathbf{P} = d\begin{pmatrix}\cos^2\phi - 1\\ -\sin\phi\cos\phi\end{pmatrix} = d\sin\phi\begin{pmatrix}-\sin\phi\\ -\cos\phi\end{pmatrix} \tag{1.2.1.4}$$

The unit direction vectors are $\mathbf{m}_1 = \begin{pmatrix}-\sin\phi\\ \cos\phi\end{pmatrix}$ and $\mathbf{m}_2 = \begin{pmatrix}-\sin\phi\\ -\cos\phi\end{pmatrix}$, so the angle between the tangents satisfies

$$\cos\angle APB = \mathbf{m}_1^\top\mathbf{m}_2 = \sin^2\phi - \cos^2\phi = -\cos 2\phi \tag{1.2.1.5}$$

Given $\angle APB = 80^\circ$,

$$\cos 2\phi = -\cos 80^\circ = \cos 100^\circ \implies \phi = 50^\circ \implies \angle POA = 50^\circ \quad \text{(Option B)} \tag{1.2.1.6}$$

Taking $r = 2$, we get $d = \frac{r}{\cos 50^\circ} \approx 3.11$ and the points

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix}3.11\\0\end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix}1.29\\1.53\end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix}1.29\\-1.53\end{pmatrix} \tag{1.2.1.7}$$

The following code verifies the solution

10/codes/1.py

The following code generates Fig. 1.2.1.1.

10/codes/1plot.py

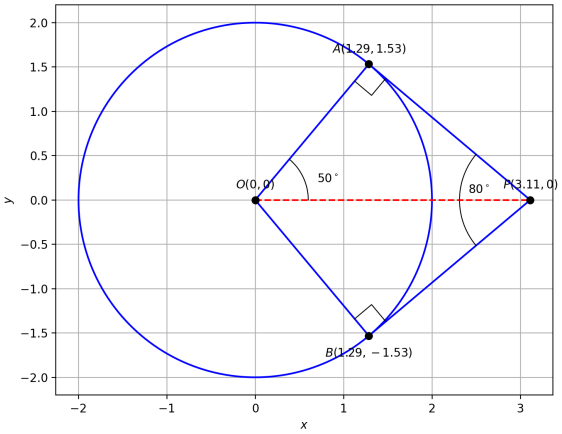


Fig. 1.2.1.1: Tangents PA and PB plotted from the coordinates of **O**, **P**, **A**, **B**

1.2.2 Two vertices of $\triangle PQR$ are $P\,(-1,5)$ and $Q\,(5,2)$. The coordinates of the point R on PQ which divides PQ in the ratio $2:1$ are (10, 2025)

a) $(3,-3)$ b) $(5,5)$ c) $(3,3)$ d) $(5,1)$

Solution: The point $\mathbf{R}$ divides the segment joining $\mathbf{P} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ internally in the ratio $m:n = 2:1$. By the section formula,

$$\mathbf{R} = \frac{m\mathbf{Q} + n\mathbf{P}}{m+n} = \frac{2\begin{pmatrix} 5 \\ 2 \end{pmatrix} + 1\begin{pmatrix} -1 \\ 5 \end{pmatrix}}{2+1} = \frac{1}{3}\begin{pmatrix} 9 \\ 9 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \quad \text{(Option C)} \quad (1.2.2.1)$$

The following code verifies the solution

10/codes/4.py

The following code generates Fig. 1.2.2.1.

10/codes/4plot.py

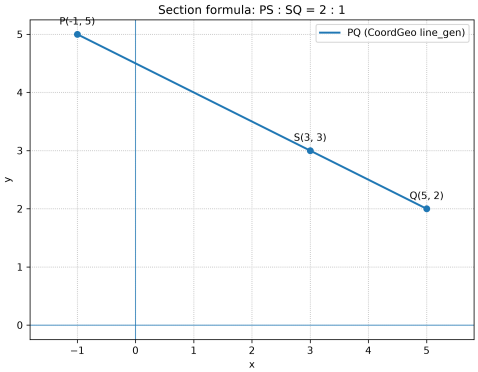


Fig. 1.2.2.1: Point R dividing PQ in the ratio $2:1$

1.2.3 If $\sin 4\theta = \frac{\sqrt{3}}{2}$, then $\frac{\theta}{3}$ is equal to (10, 2025)

a) 60° b) 20° c) 15° d) 5°

Solution: Taking the principal (acute) value,

$$4\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ \implies \theta = 15^\circ \implies \frac{\theta}{3} = 5^\circ \quad \text{(Option D)} \tag{1.2.3.1}$$

The following code verifies the solution

10/codes/5.py

1.2.4 The zeroes of the polynomial $p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3}$ are (10, 2025)

a) $-\frac{2}{3}, -\frac{1}{7}$ b) $-\frac{2}{7}, -\frac{1}{3}$ c) $\frac{2}{3}, \frac{1}{7}$ d) $\frac{2}{3}, -\frac{1}{7}$

Solution: Multiplying by 3, the zeroes of p are the zeroes of

$$g(y) = 21y^2 - 11y - 2, \quad g'(y) = 42y - 11 \tag{1.2.4.1}$$

To find a zero iteratively, expand g about the current estimate y_n to first order and require the next estimate y_{n+1} to be a zero,

$$g(y_{n+1}) \approx g(y_n) + g'(y_n)(y_{n+1} - y_n) = 0 \tag{1.2.4.2}$$

which gives the difference equation (Newton–Raphson iteration)

$$y_{n+1} = y_n - \frac{g(y_n)}{g'(y_n)} = y_n - \frac{21y_n^2 - 11y_n - 2}{42y_n - 11} \tag{1.2.4.3}$$

with the initial value y_0. Starting from $y_0 = 0.7$ and $y_0 = -0.2$, the iterations converge to

$$y = \frac{2}{3}, \quad y = -\frac{1}{7} \quad \text{(Option D)} \tag{1.2.4.4}$$

which agree with the factorisation $g(y) = (3y-2)(7y+1)$. The following code verifies the solution

10/codes/6.c

The following code generates Fig. 1.2.4.1.

10/codes/6plot.py

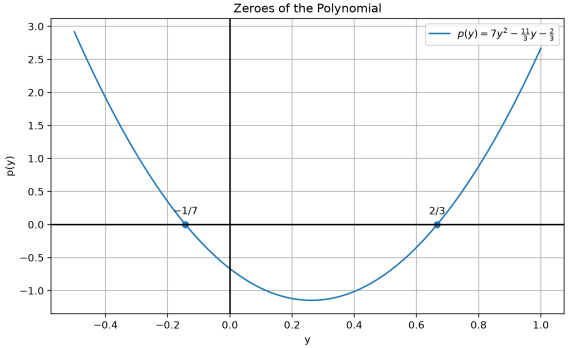


Fig. 1.2.4.1: Graph of $p(y)$ and its zeroes

1.2.5 $AOBC$ is a rectangle whose three vertices are $A(0,2)$, $O(0,0)$ and $B(4,0)$. The square of the length of its diagonal is (10, 2025)

a) 36 b) 20 c) 16 d) 4

Solution: The fourth vertex is $C(4,2)$ and the diagonal is AB. With $\mathbf{A} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$,

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}, \qquad \|\mathbf{B} - \mathbf{A}\|^2 = (\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A}) = 16 + 4 = 20 \quad \text{(Option B)} \tag{1.2.5.1}$$

The following code verifies the solution

10/codes/8.py

The following code generates Fig. 1.2.5.1.

10/codes/8plot.py

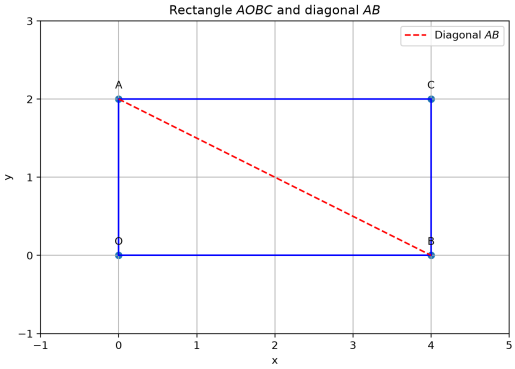


Fig. 1.2.5.1: Rectangle $AOBC$ and its diagonals

1.2.6 A die is thrown once. The probability of getting a number which is not a factor of 36 is (10, 2025)

a) $\frac{1}{2}$ b) $\frac{2}{3}$ c) $\frac{1}{6}$ d) $\frac{5}{6}$

Solution: Let X be the number on the upper face of a fair die. Its pmf is

x	1	2	3	4	5	6
$p_X(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

TABLE 1.2.6: pmf of the number on a fair die

Since $36 = 1 \times 36 = 2 \times 18 = 3 \times 12 = 4 \times 9 = 6 \times 6$, the numbers $1, 2, 3, 4, 6$ are factors of 36 and the event E ("not a factor of 36") is $E = \{5\}$. The probability of an event is the sum of the pmf over its outcomes,

$$\Pr(E) = \sum_{x \in E} p_X(x) = p_X(5) = \frac{1}{6} \quad \text{(Option C)} \tag{1.2.6.1}$$

The following program simulates 10^6 throws of the die and compares the relative frequencies with the pmf and with $\Pr(E)$

10/codes/10.c

The following code makes a stem plot of the simulated relative frequencies and the pmf, see Fig. 1.2.6.1.

10/codes/10plot.py

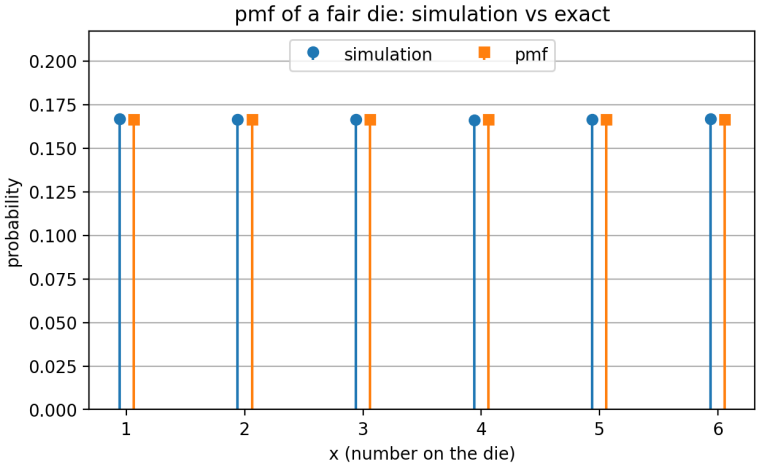


Fig. 1.2.6.1: Stem plot of the simulated relative frequencies and the pmf of a fair die

1.2.7 In the figure, $\angle BAC = 90^\circ$ and $AD \perp BC$. If $BC = 16$ cm and $DC = 4$ cm, then $AC = x$ is equal to (10, 2025)

a) 4 cm b) 5 cm c) 8 cm d) 3 cm

Solution: Take $\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} 16 \\ 0 \end{pmatrix}$. Then $BD = 16 - 4 = 12$, so $\mathbf{D} = \begin{pmatrix} 12 \\ 0 \end{pmatrix}$ and, since $AD \perp BC$, $\mathbf{A} = \begin{pmatrix} 12 \\ h \end{pmatrix}$. As $\angle BAC = 90^\circ$,

$$(\mathbf{B} - \mathbf{A})^\top (\mathbf{C} - \mathbf{A}) = 0 \tag{1.2.7.1}$$

$$\begin{pmatrix} -12 & -h \end{pmatrix} \begin{pmatrix} 4 \\ -h \end{pmatrix} = -48 + h^2 = 0 \implies h^2 = 48 \tag{1.2.7.2}$$

Hence,

$$x^2 = \|\mathbf{C} - \mathbf{A}\|^2 = 4^2 + h^2 = 64 \implies x = 8 \text{ cm} \quad \text{(Option C)} \tag{1.2.7.3}$$

The following code verifies the solution

10/codes/11.py

The following code generates Fig. 1.2.7.1.

10/codes/11plot.py

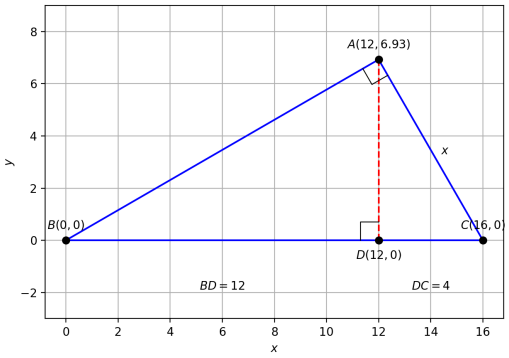


Fig. 1.2.7.1: Right triangle ABC with altitude AD, plotted from the coordinates of **B**, **C**, **D**, **A**

1.2.8 The mid-point of the line segment joining $(a, 4)$ and $(2, 2b)$ is $(2, 6)$. Then $a + b$ is equal to (10, 2025)

a) 6 b) 7 c) 8 d) 16

Solution: Let $\mathbf{P} = \begin{pmatrix} a \\ 4 \end{pmatrix}$, $\mathbf{Q} = \begin{pmatrix} 2 \\ 2b \end{pmatrix}$ and let $\mathbf{M} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$ be the mid-point. The mid-point is the average of the end points, so

$$\mathbf{M} = \frac{\mathbf{P} + \mathbf{Q}}{2} \implies 2\mathbf{M} = \mathbf{P} + \mathbf{Q} = \begin{pmatrix} a + 2 \\ 4 + 2b \end{pmatrix} \tag{1.2.8.1}$$

Separating the known entries from the unknown ones,

$$2\mathbf{M} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} a \\ 2b \end{pmatrix} \tag{1.2.8.2}$$

Let $\mathbf{C} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$ be the known part. Then

$$\begin{pmatrix} a \\ 2b \end{pmatrix} = 2\mathbf{M} - \mathbf{C} \tag{1.2.8.3}$$

The unknown vector can be written with the unknowns a and b themselves,

$$\begin{pmatrix} a \\ 2b \end{pmatrix} = \mathbf{D} \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \tag{1.2.8.4}$$

so that $\mathbf{D} \begin{pmatrix} a \\ b \end{pmatrix} = 2\mathbf{M} - \mathbf{C}$. The matrix $\mathbf{D}$ is diagonal with non-zero entries, hence invertible, and

$$\begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{D}^{-1} \left(2\mathbf{M} - \mathbf{C} \right), \qquad \mathbf{D}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \tag{1.2.8.5}$$

We need $a+b=\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix}$, therefore

$$a+b=\begin{pmatrix}1 & 1\end{pmatrix}\mathbf{D}^{-1}\left(2\mathbf{M}-\mathbf{C}\right) \tag{1.2.8.6}$$

Substituting $2\mathbf{M}-\mathbf{C}=\begin{pmatrix}4\\12\end{pmatrix}-\begin{pmatrix}2\\4\end{pmatrix}=\begin{pmatrix}2\\8\end{pmatrix}$,

$$\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}1 & 0\\0 & \frac{1}{2}\end{pmatrix}\begin{pmatrix}2\\8\end{pmatrix}=\begin{pmatrix}2\\4\end{pmatrix}\implies a+b=\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}2\\4\end{pmatrix}=6 \quad \text{(Option A)} \tag{1.2.8.7}$$

Check: for $\mathbf{P}=\begin{pmatrix}2\\4\end{pmatrix}$ and $\mathbf{Q}=\begin{pmatrix}2\\8\end{pmatrix}$ the mid-point is $\frac{1}{2}\begin{pmatrix}4\\12\end{pmatrix}=\begin{pmatrix}2\\6\end{pmatrix}$. The following code verifies the solution

10/codes/15.py

The following code generates Fig. 1.2.8.1.

10/codes/15plot.py

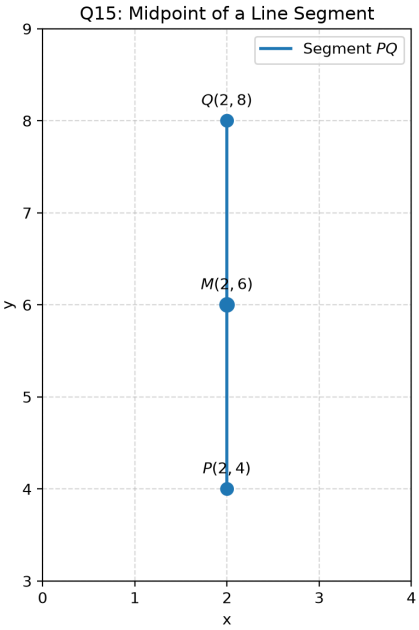


Fig. 1.2.8.1: Segment PQ and its mid-point M

1.2.9 If α and β are the zeroes of the polynomial x^2-2x-1, find the value of $\frac{1}{2\alpha}+\frac{1}{2\beta}+3\alpha\beta$. (10, 2025)

Solution: Let $f(x) = x^2 - 2x - 1$, so $f'(x) = 2x - 2$. Expanding f about the current estimate x_n to first order and requiring $f(x_{n+1}) = 0$,

$$f(x_n) + f'(x_n)(x_{n+1} - x_n) = 0 \tag{1.2.9.1}$$

gives the difference equation (Newton–Raphson iteration)

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2x_n - 1}{2x_n - 2} \tag{1.2.9.2}$$

Starting from $x_0 = 2.5$ and $x_0 = -0.5$ gives

$$\alpha = 1 + \sqrt{2}, \quad \beta = 1 - \sqrt{2} \tag{1.2.9.3}$$

From the sum and product of zeroes, $\alpha + \beta = 2$ and $\alpha\beta = -1$, so

$$\frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta = \frac{\alpha + \beta}{2\alpha\beta} + 3\alpha\beta = \frac{2}{-2} - 3 = -4 \tag{1.2.9.4}$$

The following code verifies the solution

10/codes/21.c

The following code generates Fig. 1.2.9.1.

10/codes/21plot.py

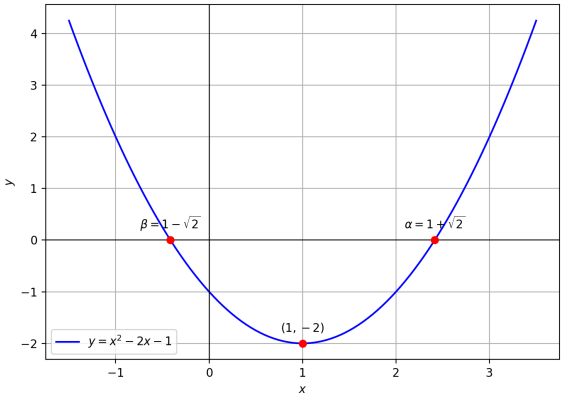


Fig. 1.2.9.1: Graph of $y = x^2 - 2x - 1$ with zeroes α and β

1.2.10 In a school, 65% of the students have black eyes, 25% have brown eyes and the rest have blue eyes. A student is chosen at random. Find the probability that the student has (a) blue eyes, (b) brown or black eyes. (10, 2025)

Solution: Let $X = 0, 1, 2$ denote black, brown and blue eyes. The probability of blue eyes is $1 - 0.65 - 0.25 = 0.10$, so the pmf is

x	0 (black)	1 (brown)	2 (blue)
$p_X(x)$	0.65	0.25	0.10

TABLE 1.2.10: pmf of the eye colour

The probability of an event is the sum of the pmf over its outcomes.
(a) Blue eyes:

$$\Pr(\text{blue}) = p_X(2) = 0.10 \tag{1.2.10.1}$$

(b) Brown or black eyes:

$$\Pr(\text{brown or black}) = p_X(1) + p_X(0) = 0.25 + 0.65 = 0.90 \tag{1.2.10.2}$$

which is also $1 - \Pr(\text{blue}) = 0.90$. The following program simulates the eye colours of 10^6 people and compares the relative frequencies with the pmf

10/codes/27.c

The following code makes a stem plot of the simulated relative frequencies and the pmf, see Fig. 1.2.10.1.

10/codes/27plot.py

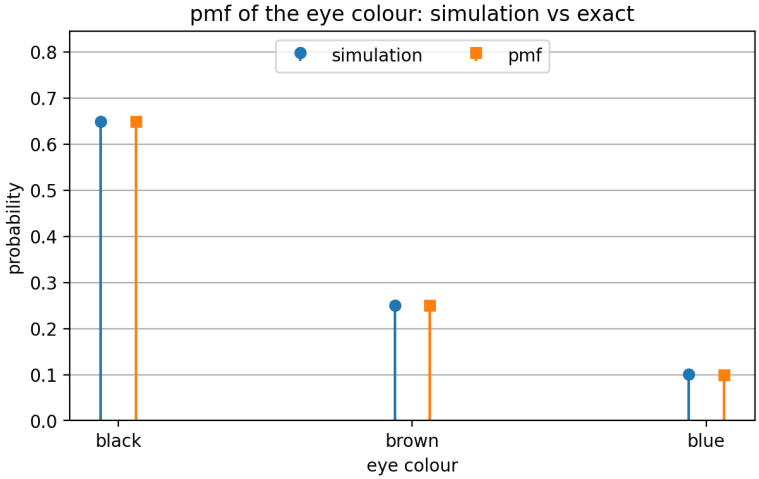


Fig. 1.2.10.1: Stem plot of the simulated relative frequencies and the pmf of the eye colour

1.2.11 In the figure, PC is a tangent to the circle at C and AB is a diameter which, when extended, meets the tangent at P. If $\angle PCA = 110°$, find $\angle CBA$ and $\angle BCO$. (10, 2025)

Solution: Since AB is a diameter, $\angle ACB = 90^\circ$. As P lies beyond B,

$$\angle PCA = \angle PCB + \angle BCA \implies \angle PCB = 110^\circ - 90^\circ = 20^\circ \tag{1.2.11.1}$$

By the tangent–chord theorem, $\angle CAB = \angle PCB = 20^\circ$. In $\triangle ABC$,

$$\angle CBA = 90^\circ - 20^\circ = 70^\circ \tag{1.2.11.2}$$

Since $OB = OC$, $\triangle OBC$ is isosceles and

$$\angle BCO = \angle CBO = 70^\circ \tag{1.2.11.3}$$

Assumptions for the coordinates. The problem fixes only angles, so the coordinate frame can be chosen freely without changing the answer:

(i) the centre is placed at the origin, $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ (a translation does not change angles);

(ii) the diameter AB is placed along the x-axis with A on the negative side (a rotation does not change angles);

(iii) the radius is taken as $r = 2$ (any other r gives a similar figure with the same angles);

(iv) the point C is taken above the x-axis (its reflection in AB gives the mirror image of the figure).

With these choices only the angle data are needed to locate $\mathbf{C}$ and $\mathbf{P}$. Then

$$\mathbf{A} = \begin{pmatrix} -r \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} r \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \tag{1.2.11.4}$$

Point C. It lies on the circle, so $\mathbf{C} = r \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$, where $\theta = \angle BOC$ and $\mathbf{C}$ is taken above the x-axis. To find θ from $\angle CAB = 20^\circ$, look at the direction of $\mathbf{AC}$. Using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$,

$$\mathbf{C} - \mathbf{A} = r \begin{pmatrix} 1 + \cos\theta \\ \sin\theta \end{pmatrix} = 2r\cos\frac{\theta}{2} \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \end{pmatrix} \tag{1.2.11.5}$$

so AC makes the angle $\frac{\theta}{2}$ with AB (the x-axis). Therefore

$$\angle CAB = \frac{\theta}{2} = 20^\circ \implies \theta = 40^\circ \implies \mathbf{C} = 2 \begin{pmatrix} \cos 40^\circ \\ \sin 40^\circ \end{pmatrix} = \begin{pmatrix} 1.53 \\ 1.29 \end{pmatrix} \tag{1.2.11.6}$$

which agrees with the fact that the central angle is twice the angle at the circumference.

Point P. The tangent at $\mathbf{C}$ is perpendicular to the radius $\mathbf{OC}$, and $\mathbf{P} = \begin{pmatrix} p \\ 0 \end{pmatrix}$ lies on it, so

$$(\mathbf{P} - \mathbf{C})^\top (\mathbf{C} - \mathbf{O}) = 0 \implies pr\cos\theta - r^2 = 0 \implies p = \frac{r}{\cos\theta} = \frac{2}{\cos 40^\circ} = 2.61 \tag{1.2.11.7}$$

Since $p > r$, the point $\mathbf{P}$ lies beyond $\mathbf{B}$ on AB produced, as assumed in the solution.

Point	**O**	**A**	**B**	**C**	**P**
Coordinates	$(0,0)$	$(-2,0)$	$(2,0)$	$(1.53,1.29)$	$(2.61,0)$

TABLE 1.2.11: Coordinates used to plot the figure

Checking the angles in the figure. For $\angle ACB$,

$$(\mathbf{A}-\mathbf{C})^{\top}(\mathbf{B}-\mathbf{C}) = -r^2(1+\cos\theta)(1-\cos\theta) + r^2\sin^2\theta = 0 \qquad (1.2.11.8)$$

so $\angle ACB = 90^\circ$, and $\angle OCP = 90^\circ$ by the choice of $\mathbf{P}$; both are marked in the figure. For $\angle PCA$, with $\mathbf{P}-\mathbf{C} = \begin{pmatrix} 1.08 \\ -1.29 \end{pmatrix}$ and $\mathbf{A}-\mathbf{C} = \begin{pmatrix} -3.53 \\ -1.29 \end{pmatrix}$,

$$\cos\angle PCA = \frac{(\mathbf{P}-\mathbf{C})^{\top}(\mathbf{A}-\mathbf{C})}{\|\mathbf{P}-\mathbf{C}\|\,\|\mathbf{A}-\mathbf{C}\|} = \frac{-2.16}{1.68\times 3.76} = -0.342 = \cos 110^\circ \qquad (1.2.11.9)$$

The graph. The final figure is obtained by plotting these five points, joining A–P (the line AB produced), P–C, A–C, C–B and O–C (dashed), and drawing the circle of radius 2 about $\mathbf{O}$ through the points $r\begin{pmatrix}\cos t \\ \sin t\end{pmatrix}$, $0 \le t \le 2\pi$. The following code verifies the solution

10/codes/29.py

The following code generates Fig. 1.2.11.1.

10/codes/29plot.py

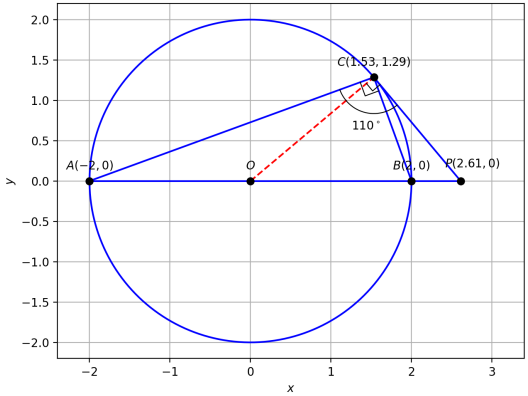


Fig. 1.2.11.1: Tangent PC and diameter AB, plotted from the coordinates of **A**, **B**, **C**, **P**

1.2.12 Find the coordinates of the points which divide the line segment joining $A\,(-2,2)$ and $B\,(2,8)$ into four equal parts. (10, 2025)

Solution: Let $\mathbf{A} = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$. The points that divide AB into four equal parts are

$$\mathbf{P}_k = \mathbf{A} + \frac{k}{4}\left(\mathbf{B} - \mathbf{A}\right), \quad k = 1, 2, 3 \tag{1.2.12.1}$$

Collecting the terms in $\mathbf{A}$ and $\mathbf{B}$,

$$\mathbf{P}_k = \left(1 - \frac{k}{4}\right)\mathbf{A} + \frac{k}{4}\mathbf{B} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} \begin{pmatrix} 1 - \frac{k}{4} \\ \frac{k}{4} \end{pmatrix} \tag{1.2.12.2}$$

The weights add up to 1, as they must for a point on the segment. Writing the weights for $k = 1, 2, 3$ side by side as the columns of one matrix,

$$\begin{pmatrix} \mathbf{P}_1 & \mathbf{P}_2 & \mathbf{P}_3 \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix} \tag{1.2.12.3}$$

and the columns of the product give the three points. Substituting $\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ 2 & 8 \end{pmatrix}$,

$$\begin{pmatrix} \mathbf{P}_1 & \mathbf{P}_2 & \mathbf{P}_3 \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ 2 & 8 \end{pmatrix} \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix} \tag{1.2.12.4}$$

$$= \begin{pmatrix} -\frac{3}{2} + \frac{1}{2} & -1 + 1 & -\frac{1}{2} + \frac{3}{2} \\ \frac{3}{2} + 2 & 1 + 4 & \frac{1}{2} + 6 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 1 \\ 3.5 & 5 & 6.5 \end{pmatrix} \tag{1.2.12.5}$$

Hence the points are

$$\mathbf{P}_1 = \begin{pmatrix} -1 \\ 3.5 \end{pmatrix}, \quad \mathbf{P}_2 = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \quad \mathbf{P}_3 = \begin{pmatrix} 1 \\ 6.5 \end{pmatrix} \tag{1.2.12.6}$$

As a check, $\mathbf{P}_2$ is the mid-point $\frac{\mathbf{A}+\mathbf{B}}{2} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$. The following code verifies the solution

10/codes/31.py

The following code generates Fig. 1.2.12.1.

10/codes/31plot.py

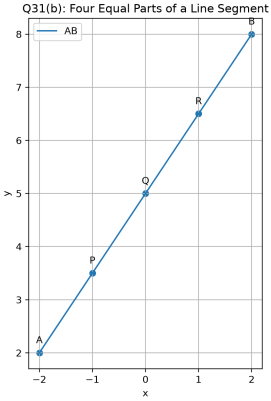


Fig. 1.2.12.1: Division of AB into four equal parts

1.2.13 The nth term of an AP is 4, its common difference is 2 and the sum of its first n terms is -14. Find n, the first term, and the sum of the first 20 terms. (10, 2025)
Solution: Let $u[n]$ be the unit step function, $u[n] = 1$ for $n \geq 0$ and 0 otherwise. Index the terms from $n = 0$, so that the AP with first term a and common difference d is the causal sequence

$$x[n] = (a + nd)\ u[n] \tag{1.2.13.1}$$

with $x[0] = a$. The running sum $y[n] = x[0] + \cdots + x[n]$ is the convolution $y = x * u$, so in the z-domain $Y(z) = X(z)\,U(z)$. Using the transform pairs

$$u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - z^{-1}}, \quad n\,u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-1}}{(1 - z^{-1})^2} \tag{1.2.13.2}$$

the transform of the AP is

$$X(z) = \frac{a}{1 - z^{-1}} + \frac{d\,z^{-1}}{(1 - z^{-1})^2} \tag{1.2.13.3}$$

$$\implies Y(z) = \frac{a}{(1 - z^{-1})^2} + \frac{d\,z^{-1}}{(1 - z^{-1})^3} \tag{1.2.13.4}$$

From the pairs

$$(n+1)\,u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{(1 - z^{-1})^2}, \quad \frac{n(n+1)}{2} u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-1}}{(1 - z^{-1})^3} \tag{1.2.13.5}$$

the inverse transform gives the sum of the first $n + 1$ terms,

$$y[n] = (n+1)\,a + \frac{n(n+1)}{2} d, \quad n \geq 0 \tag{1.2.13.6}$$

Let m be the index of the last term, so that the number of terms is $m + 1$. Here $d = 2$, $x[m] = 4$ and $y[m] = -14$. From $x[m] = a + 2m = 4$ we get $a = 4 - 2m$,

and

$$y[m] = (m+1)(4-2m) + m(m+1) = (m+1)(4-m) = -14 \quad (1.2.13.7)$$

$$\implies m^2 - 3m - 18 = 0 \implies (m-6)(m+3) = 0 \quad (1.2.13.8)$$

Since $m \geq 0$, $m = 6$, so the number of terms is $m+1 = 7$ and $a = 4 - 12 = -8$. The sum of the first 20 terms is $y[19]$,

$$y[19] = 20(-8) + \frac{19 \cdot 20}{2}(2) = -160 + 380 = 220 \quad (1.2.13.9)$$

The following code verifies the solution

10/codes/32a.py

1.2.14 A man lent part of his money at 10% per annum and the rest at 15% per annum. His yearly interest was ₹1900. Had he interchanged the two rates, he would have earned ₹200 more. Find the two amounts. (10, 2025)

Solution: Let x and y be the amounts lent at 10% and 15%. Then

$$0.10x + 0.15y = 1900 \quad (1.2.14.1)$$

$$0.15x + 0.10y = 2100 \quad (1.2.14.2)$$

The augmented matrix of $\mathbf{Ax} = \mathbf{b}$ is $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$. Using Gaussian elimination (Gauss–Jordan row reduction),

$$\begin{pmatrix} 0.10 & 0.15 & 1900 \\ 0.15 & 0.10 & 2100 \end{pmatrix} \xleftrightarrow{R_1 \leftarrow 10R_1} \begin{pmatrix} 1 & 1.5 & 19000 \\ 0.15 & 0.10 & 2100 \end{pmatrix} \quad (1.2.14.3)$$

$$\xleftrightarrow{R_2 \leftarrow R_2 - 0.15R_1} \begin{pmatrix} 1 & 1.5 & 19000 \\ 0 & -0.125 & -750 \end{pmatrix} \quad (1.2.14.4)$$

$$\xleftrightarrow{R_2 \leftarrow R_2/(-0.125)} \begin{pmatrix} 1 & 1.5 & 19000 \\ 0 & 1 & 6000 \end{pmatrix} \quad (1.2.14.5)$$

$$\xleftrightarrow{R_1 \leftarrow R_1 - 1.5R_2} \begin{pmatrix} 1 & 0 & 10000 \\ 0 & 1 & 6000 \end{pmatrix} \quad (1.2.14.6)$$

Both $\mathbf{A}$ and $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$ have two non-zero rows, so

$$\text{rank}(\mathbf{A}) = \text{rank}\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix} = 2 = \text{number of unknowns} \quad (1.2.14.7)$$

Hence the system is consistent with a unique solution,

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10000 \\ 6000 \end{pmatrix} \quad (1.2.14.8)$$

The amounts are ₹10000 at 10% and ₹6000 at 15%. The following code verifies the solution

10/codes/34.py

1.2.15 **Case study (lighthouse).** A lighthouse AB is 50 m high. From its top B, the angles of depression of a ship at P and, after some time, at Q are 30° and 45° respectively, with P, Q, A on the same line. (10, 2025)

Solution: Take the foot of the lighthouse as the origin, $\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 \\ 50 \end{pmatrix}$, with the ground along the x-axis, $\mathbf{e}_2^\top \mathbf{x} = 0$. The line of sight from $\mathbf{B}$ with angle of depression θ has direction vector $\mathbf{m} = \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix}$ and meets the ground when

$$\mathbf{e}_2^\top (\mathbf{B} + \lambda \mathbf{m}) = 0 \implies \lambda = -\frac{\mathbf{e}_2^\top \mathbf{B}}{\mathbf{e}_2^\top \mathbf{m}} = \frac{50}{\sin\theta} \tag{1.2.15.1}$$

For $\mathbf{Q}$ ($\theta = 45^\circ$) and $\mathbf{P}$ ($\theta = 30^\circ$),

$$\mathbf{Q} = \mathbf{B} + 50\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 50 \\ 0 \end{pmatrix} \tag{1.2.15.2}$$

$$\mathbf{P} = \mathbf{B} + 100 \cdot \frac{1}{2} \begin{pmatrix} \sqrt{3} \\ -1 \end{pmatrix} = \begin{pmatrix} 50\sqrt{3} \\ 0 \end{pmatrix} \tag{1.2.15.3}$$

(i) The distance QA is

$$QA = \|\mathbf{Q} - \mathbf{A}\| = 50 \text{ m} \tag{1.2.15.4}$$

(iii)(a) The distance moved from P to Q is

$$PQ = \|\mathbf{P} - \mathbf{Q}\| = 50\left(\sqrt{3} - 1\right) \approx 36.60 \text{ m} \tag{1.2.15.5}$$

(iii)(b) The ship covers $QA = 50$ m $= 0.05$ km in 10 minutes $= \frac{1}{6}$ h, so

$$\text{Speed} = \frac{0.05}{1/6} = 0.3 \text{ km/h} \tag{1.2.15.6}$$

The following code verifies the solution

10/codes/38.py

The following code generates Fig. 1.2.15.1.

10/codes/38plot.py

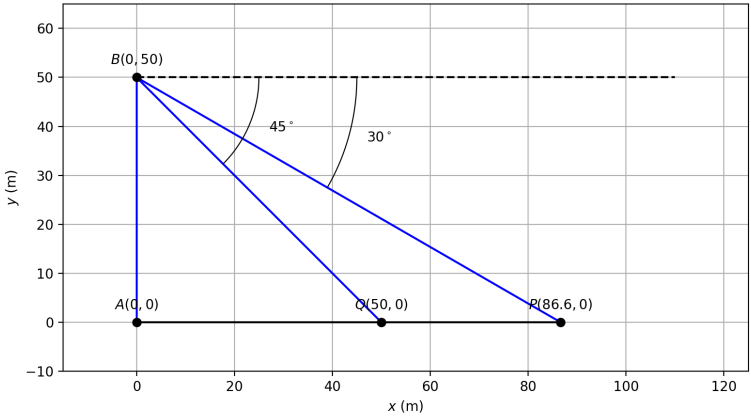


Fig. 1.2.15.1: Lighthouse AB and ship positions P, Q, plotted from coordinates

2. 12

A. *Formulae*

2.1.1 Domain of inverse functions. $\cos^{-1} x$ and $\sin^{-1} x$ are defined only for $-1 \leq x \leq 1$, while $\sin x$ and $\cos x$ are defined for all real x. The domain of a sum is the intersection of the domains.

2.1.2 Counting. If each of m independent positions can take c values, the total number of arrangements is

$$N = c^m \tag{2.1.2.1}$$

2.1.3 Symmetric matrix. A square matrix is symmetric if

$$\mathbf{A}^\top = \mathbf{A} \tag{2.1.3.1}$$

Every diagonal matrix is symmetric.

2.1.4 Determinant properties. For $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and a scalar k,

$$|k\mathbf{A}| = k^n |\mathbf{A}|, \qquad |\mathbf{AB}| = |\mathbf{A}|\,|\mathbf{B}| \tag{2.1.4.1}$$

2.1.5 Adjugate. With $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$,

$$|\mathrm{adj}\,\mathbf{A}| = |\mathbf{A}|^{n-1} \tag{2.1.5.1}$$

Solution: Taking determinants of $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$ gives $|\mathbf{A}|\,|\mathrm{adj}\,\mathbf{A}| = |\mathbf{A}|^n$.

2.1.6 Characteristic polynomial and Cayley–Hamilton theorem. The characteristic polynomial of a square matrix is $f(\lambda) = |\mathbf{A} - \lambda\mathbf{I}|$, and

$$|\mathbf{A}| = f(0) = \lambda_1 \cdots \lambda_n, \qquad f(\mathbf{A}) = \mathbf{0} \tag{2.1.6.1}$$

For a 2×2 matrix, $\mathbf{A}^2 - \mathrm{tr}(\mathbf{A})\,\mathbf{A} + |\mathbf{A}|\,\mathbf{I} = \mathbf{0}$; the eigenvalues sum to the trace and multiply to the determinant. For a 3×3 matrix with $f(\lambda) = -\lambda^3 + s_1\lambda^2 - s_2\lambda + s_3$,

Cayley–Hamilton gives

$$\operatorname{adj}\mathbf{A} = \mathbf{A}^2 - s_1\mathbf{A} + s_2\mathbf{I} \tag{2.1.6.2}$$

2.1.7 Gaussian elimination and rank. For $\mathbf{Ax} = \mathbf{b}$ with n unknowns, row-reduce the augmented matrix $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$. The system is consistent if and only if

$$\operatorname{rank}(\mathbf{A}) = \operatorname{rank}\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix} \tag{2.1.7.1}$$

and the solution is unique when this common rank equals n.

2.1.8 Monotonicity. If $f'(x) > 0$ for all x in an interval, f is strictly increasing there; if $f'(x) < 0$, it is strictly decreasing.

2.1.9 Trapezoidal rule. With n subintervals of width $h = \frac{b-a}{n}$ and nodes $x_i = a + ih$,

$$\int_a^b f(x)\,dx \approx T_n = \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right) \tag{2.1.9.1}$$

The error is $-\frac{(b-a)h^2}{12}f''(\xi)$ for some $\xi \in (a,b)$, so T_n converges to the integral as $n \to \infty$.

2.1.10 Trapezoidal rule as a difference equation. With $I[n] = \int_a^{x_n} f(x)\,dx$ and $I[0] = 0$,

$$I[n+1] = I[n] + \frac{h}{2}\left(f(x_n) + f(x_{n+1})\right) \tag{2.1.10.1}$$

and the integral is $I[N]$.

2.1.11 Derivatives. Product rule, chain rule and the exponential rule,

$$\frac{d}{dx}(uv) = u'v + uv', \quad \frac{d}{dx}a^x = a^x \ln a, \quad \frac{d}{dx}\log(\sin u) = \cot u\,\frac{du}{dx} \tag{2.1.11.1}$$

2.1.12 Symmetric integral. If $g(\pi - x) = g(x)$,

$$\int_0^\pi x\,g(x)\,dx = \frac{\pi}{2}\int_0^\pi g(x)\,dx \tag{2.1.12.1}$$

Solution: Substituting $x \to \pi - x$ gives $I = \int_0^\pi (\pi - x)\,g(x)\,dx$. Adding this to the original, $2I = \pi\int_0^\pi g(x)\,dx$.

2.1.13 Laplace transform. For a first-order equation $y' + ay = f(x)$,

$$\mathcal{L}\{y'\} = sY(s) - y(0), \quad \mathcal{L}\{\cos x\} = \frac{s}{s^2+1}, \quad \mathcal{L}\left\{e^{-ax}\right\} = \frac{1}{s+a} \tag{2.1.13.1}$$

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t\} = \frac{1}{s^2} \tag{2.1.13.2}$$

The order of a differential equation is the highest derivative present; the degree is its power.

2.1.14 Orthogonality and scalar product with a unit vector,

$$|\mathbf{a}+\mathbf{b}| = |\mathbf{a}-\mathbf{b}| \iff \mathbf{a}^\top\mathbf{b} = 0, \qquad \text{scalar product of } \mathbf{a} \text{ with unit vector along } \mathbf{v} = \frac{\mathbf{a}^\top}{\|\mathbf{v}}$$

(2.1.14.1)

2.1.15 Shortest distance between lines $\mathbf{x} = \mathbf{A} + \kappa_1 \mathbf{m}_1$ and $\mathbf{x} = \mathbf{B} + \kappa_2 \mathbf{m}_2$. If $\operatorname{rank}\begin{pmatrix}\mathbf{m}_1 & \mathbf{m}_2\end{pmatrix} = 1$ the lines are parallel and the distance is the least value of $\|\mathbf{A} + \kappa \mathbf{m}_1 - \mathbf{B}\|$, obtained from the normal equation

$$\mathbf{m}_1^\top \mathbf{m}_1 \, \kappa = \mathbf{m}_1^\top (\mathbf{B} - \mathbf{A}) \tag{2.1.15.1}$$

For skew lines, $\mathbf{M} = \begin{pmatrix}\mathbf{m}_1 & \mathbf{m}_2\end{pmatrix}$ and the least squares solution of $\mathbf{M}\kappa = \mathbf{B} - \mathbf{A}$ is used.

2.1.16 Points on a line in space. A line $\mathbf{x} = \mathbf{A} + \kappa \mathbf{m}$ is written with one parameter κ, and a point of it is at distance r from $\mathbf{P}$ when $\|\mathbf{A} + \kappa \mathbf{m} - \mathbf{P}\|^2 = r^2$, a quadratic in κ.

2.1.17 Foot of the perpendicular and image. For a point $\mathbf{P}$ and the line $\mathbf{A} + \kappa \mathbf{m}$,

$$\kappa_0 = \frac{\mathbf{m}^\top (\mathbf{P} - \mathbf{A})}{\mathbf{m}^\top \mathbf{m}}, \quad \mathbf{H} = \mathbf{A} + \kappa_0 \mathbf{m}, \quad \mathbf{P}' = 2\mathbf{H} - \mathbf{P} \tag{2.1.17.1}$$

2.1.18 Corner point theorem. The maximum or minimum of a linear objective over a feasible region occurs at a corner point. For an unbounded region, a minimum exists only if the objective is bounded below there.

2.1.19 Linear programming in standard form. "Maximise $\mathbf{c}^\top \mathbf{x}$ subject to $\mathbf{A}\mathbf{x} \preceq \mathbf{b}$, $\mathbf{x} \succeq \mathbf{0}$" (or minimise with $\succeq$) is solved numerically with `cvxpy`.

2.1.20 Conditional probability,

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \qquad \Pr(A' \cap B) = \Pr(B) - \Pr(A \cap B) \tag{2.1.20.1}$$

and $\Pr(A \cap B) \le \min\{\Pr(A), \Pr(B)\}$.

2.1.21 Probability from a pmf. For a random variable X with pmf $p_X(x)$ and an event E,

$$\Pr(E) = \sum_{x \in E} p_X(x), \quad \sum_x p_X(x) = 1, \quad E[X] = \sum_x x \, p_X(x) \tag{2.1.21.1}$$

For independent X and Y the joint pmf is $p_{XY}(x, y) = p_X(x) \, p_Y(y)$.

2.1.22 Total probability and Bayes' theorem. For mutually exclusive, exhaustive events $A_1, \ldots, A_n$,

$$\Pr(D) = \sum_i \Pr(D \mid A_i) \Pr(A_i), \qquad \Pr(A_1 \mid D) = \frac{\Pr(D \mid A_1) \Pr(A_1)}{\Pr(D)} \tag{2.1.22.1}$$

B. Problems

2.2.1 Find the domain of the function $f(x) = \cos^{-1} x + \sin x$. (12, 2025)

Solution: $\cos^{-1} x$ is defined only for $-1 \le x \le 1$, and $\sin x$ is defined for every real x. The domain of the sum is the intersection of the two domains:

$$\text{Domain}(f) = [-1, 1] \quad \text{(Option C)} \tag{2.2.1.1}$$

The program scans x over $[-3, 3]$ and treats the points where `acos(x)+sin(x)` is not a number as outside the domain. The following code verifies the solution

12/codes/2.py

2.2.2 The number of all possible matrices of order 3×3 with each entry 2 or 3 is (12, 2025)

Solution: Each of the 9 entries can independently be 2 or 3. By the multiplication principle,

$$N = 2^9 = 512 \quad \text{(Option B)} \tag{2.2.2.1}$$

The following code generates every 3×3 matrix with entries 2 or 3 and counts them in the loop

12/codes/3.py

2.2.3 The matrix $\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ is a (12, 2025)

a) scalar matrix b) identity matrix c) null matrix d) symmetric matrix

Solution: (theory only) A matrix is symmetric if $\mathbf{A}^\top = \mathbf{A}$. For a diagonal matrix the transpose leaves every entry unchanged,

$$\mathbf{A}^\top = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \mathbf{A} \tag{2.2.3.1}$$

The diagonal entries are not all equal, so $\mathbf{A}$ is not a scalar matrix; it is neither the identity nor the null matrix. Hence $\mathbf{A}$ is a symmetric matrix (Option D).

2.2.4 If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of order 3 with $|\mathbf{A}| = 3$ and $|\mathbf{B}| = 5$, then $|2\mathbf{AB}|$ is (12, 2025)

Solution: Following matgeo, the characteristic polynomial of a square matrix $\mathbf{M}$ is $f(\lambda) = |\mathbf{M} - \lambda\mathbf{I}|$, so

$$|\mathbf{M}| = f(0) = \lambda_1\lambda_2\lambda_3 \tag{2.2.4.1}$$

where λ_i are the eigenvalues. The eigenvalues of $2\mathbf{AB}$ are not those of $\mathbf{A}$ and $\mathbf{B}$ separately, so we use $|k\mathbf{A}| = k^n |\mathbf{A}|$ and $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$. With $n = 3$,

$$|2\mathbf{AB}| = 2^3 |\mathbf{A}| |\mathbf{B}| = 8(3)(5) = 120 \quad \text{(Option B)} \tag{2.2.4.2}$$

To check this numerically, take $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$ with $|\mathbf{A}| = 3$ and $\mathbf{B} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$ with $|\mathbf{B}| = 5$. The program obtains the characteristic polynomial of $\mathbf{M} = 2\mathbf{AB}$ from `numpy.poly`, which returns $\det(\lambda\mathbf{I} - \mathbf{M})$, and reads $|\mathbf{M}| = (-1)^3 c_0$ from its constant term c_0. The following code verifies the solution

12/codes/5.py

2.2.5 If $\mathbf{A}$ is a square matrix of order 3 with $|\mathbf{A}| = 5$, then $|\text{adj}\,\mathbf{A}|$ is (12, 2025)

Solution: For a square matrix of order 3,

$$\mathbf{A}\,\text{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I} \tag{2.2.5.1}$$

Taking the determinant of both sides and using $|k\mathbf{I}| = k^3$ for order 3,

$$|\mathbf{A}|\,|\text{adj}\,\mathbf{A}| = |\mathbf{A}|^3 \implies |\text{adj}\,\mathbf{A}| = |\mathbf{A}|^2 = 5^2 = 25 \quad \text{(Option C)} \tag{2.2.5.2}$$

since $|\mathbf{A}| = 5 \neq 0$. The program builds $\text{adj}\,\mathbf{A}$ from the cofactors of an example matrix with $|\mathbf{A}| = 5$ and checks that $\mathbf{A}\,\text{adj}\,\mathbf{A} = 5\mathbf{I}$ and $|\text{adj}\,\mathbf{A}| = 25$. The following code verifies the solution

12/codes/6.py

2.2.6 The function $f(x) = 2x - \sin x$, $x \in \mathbb{R}$, is (12, 2025)

a) strictly decreasing on $\mathbb{R}$
b) strictly increasing on $\mathbb{R}$
c) neither increasing nor decreasing on $\mathbb{R}$
d) decreasing on $(-\infty, 0)$ and increasing on $(0, \infty)$

Solution: Differentiating,

$$f'(x) = 2 - \cos x \tag{2.2.6.1}$$

Since $-1 \leq \cos x \leq 1$,

$$1 \leq f'(x) \leq 3 \implies f'(x) > 0 \quad \forall x \in \mathbb{R} \tag{2.2.6.2}$$

A differentiable function with a strictly positive derivative is strictly increasing, so f is strictly increasing on $\mathbb{R}$ (Option B). The program checks that $f'(x) > 0$ at every point of a fine grid. The following code verifies the solution

12/codes/11.py

The following code generates Fig. 2.2.6.1.

12/codes/11plot.py

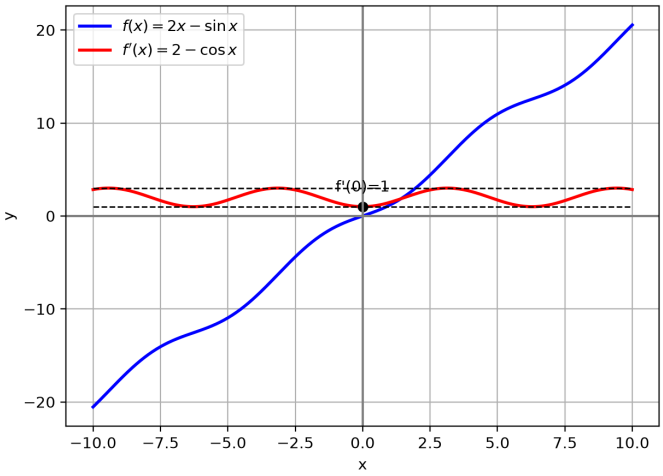


Fig. 2.2.6.1: $f(x)$ and $f'(x)$ with $1 \leq f'(x) \leq 3$

2.2.7 If $\mathbf{a}$ and $\mathbf{b}$ are two vectors such that $|\mathbf{a}+\mathbf{b}| = |\mathbf{a}-\mathbf{b}|$, then $\mathbf{a}$ and $\mathbf{b}$ are (12, 2025)

Solution: (theory only) Squaring both sides and using $|\mathbf{x}|^2 = \mathbf{x}^\top \mathbf{x}$,

$$|\mathbf{a}|^2 + 2\mathbf{a}^\top\mathbf{b} + |\mathbf{b}|^2 = |\mathbf{a}|^2 - 2\mathbf{a}^\top\mathbf{b} + |\mathbf{b}|^2 \tag{2.2.7.1}$$

$$\implies 4\mathbf{a}^\top\mathbf{b} = 0 \tag{2.2.7.2}$$

Hence $\mathbf{a}$ and $\mathbf{b}$ are orthogonal (Option A); the zero vector is the trivial case.

2.2.8 If $\Pr(A) = \frac{1}{7}$, $\Pr(B) = \frac{5}{7}$ and $\Pr(A \cap B) = \frac{4}{7}$, find $\Pr(A' \mid B)$. (12, 2025)

Solution: Let X take the values 0, 1, 2 for the outcomes $A \cap B$, $A' \cap B$ and B' respectively. From $\Pr(B) = \frac{5}{7}$ and $\Pr(A \cap B) = \frac{4}{7}$,

x	0 $(A \cap B)$	1 $(A' \cap B)$	2 (B')
$p_X(x)$	$\frac{4}{7}$	$\frac{5}{7} - \frac{4}{7} = \frac{1}{7}$	$1 - \frac{5}{7} = \frac{2}{7}$

TABLE 2.2.8: pmf of X

Then $\Pr(B) = p_X(0) + p_X(1) = \frac{5}{7}$ and $\Pr(A' \cap B) = p_X(1) = \frac{1}{7}$, so

$$\Pr(A' \mid B) = \frac{\Pr(A' \cap B)}{\Pr(B)} = \frac{1/7}{5/7} = \frac{1}{5} \tag{2.2.8.1}$$

The values as printed are not consistent with each other, because an intersection cannot be more probable than A alone ($\frac{4}{7} > \frac{1}{7}$); the answer above uses only $\Pr(B)$ and $\Pr(A \cap B)$. The following program simulates 10^6 draws from this pmf

12/codes/17.c

The following code makes a stem plot of the simulated relative frequencies and the pmf, see Fig. 2.2.8.1.

12/codes/17plot.py

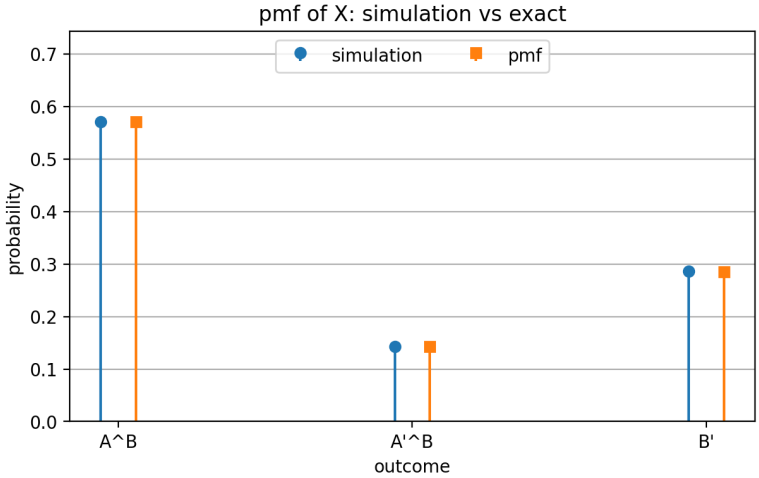


Fig. 2.2.8.1: Stem plot of the simulated relative frequencies and the pmf of X

2.2.9 A coin is tossed and a card is selected at random from a well-shuffled pack of 52 cards. The probability of getting a head and a face card is (12, 2025)
Solution: Let $X = 1$ if the coin shows a head and 0 otherwise, and $Y = 1$ if the card is a face card and 0 otherwise. There are 12 face cards among 52, so

x	0	1
$p_X(x)$	$\frac{1}{2}$	$\frac{1}{2}$
$p_Y(x)$	$\frac{40}{52} = \frac{10}{13}$	$\frac{12}{52} = \frac{3}{13}$

TABLE 2.2.9: pmfs of X (coin) and Y (card)

The coin and the card are independent, so the joint pmf is the product $p_{XY}(x,y) = p_X(x)\,p_Y(y)$. A head and a face card is the outcome $(X,Y) = (1,1)$:

$$p_{XY}(1,1) = p_X(1)\,p_Y(1) = \frac{1}{2}\cdot\frac{3}{13} = \frac{3}{26} \quad \text{(Option B)} \tag{2.2.9.1}$$

The following program simulates 10^6 tosses and draws and compares the relative frequency with this value

12/codes/18.c

The following code makes a stem plot of the simulated relative frequencies and the pmf, see Fig. 2.2.9.1.

12/codes/18plot.py

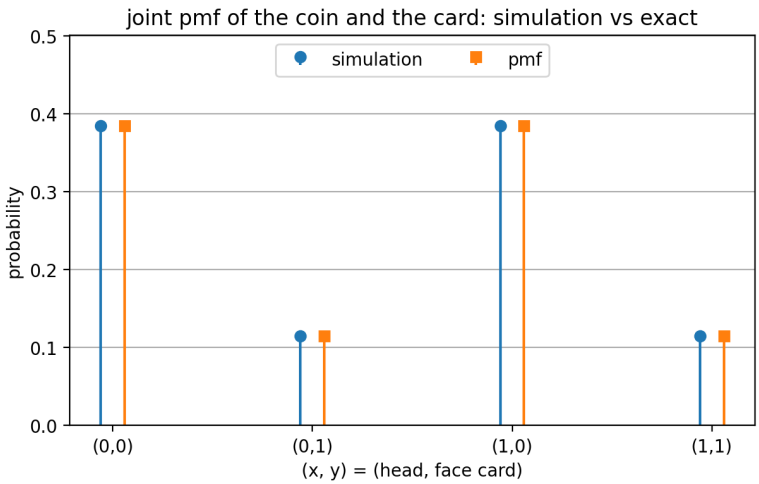


Fig. 2.2.9.1: Stem plot of the simulated relative frequencies and the joint pmf of the coin and the card

2.2.10 If $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix}$, show that $\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \mathbf{0}$. (12, 2025)

Solution: The trace and determinant are

$$\operatorname{tr}(\mathbf{A}) = 2 + 2 = 4, \qquad |\mathbf{A}| = 2(2) - 3(-1) = 7 \tag{2.2.10.1}$$

The characteristic polynomial of a 2×2 matrix is $\lambda^2 - \operatorname{tr}(\mathbf{A})\lambda + |\mathbf{A}|$, so

$$\lambda^2 - 4\lambda + 7 = 0 \implies \lambda = \frac{4 \pm \sqrt{16 - 28}}{2} = 2 \pm i\sqrt{3} \tag{2.2.10.2}$$

The eigenvalues sum to 4 (the trace) and multiply to 7 (the determinant). By the Cayley–Hamilton theorem, $\mathbf{A}$ satisfies its own characteristic equation:

$$\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \mathbf{0} \tag{2.2.10.3}$$

Direct verification: $\mathbf{A}^2 = \begin{pmatrix} 1 & 12 \\ -4 & 1 \end{pmatrix}$, $4\mathbf{A} = \begin{pmatrix} 8 & 12 \\ -4 & 8 \end{pmatrix}$, $7\mathbf{I} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$, so

$$\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \begin{pmatrix} 1 - 8 + 7 & 12 - 12 \\ -4 + 4 & 1 - 8 + 7 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \tag{2.2.10.4}$$

The following code verifies the solution

12/codes/22.py

2.2.11 Maximise $Z = 5x + 4y$ subject to $3x + y \leq 6$, $x \leq 1$, $x \geq 0$, $y \geq 0$. (12, 2025)

Solution: Step 1: the feasible region is convex. Each constraint is a half-plane $\mathbf{a}^\top \mathbf{x} \le c$, and a half-plane is a convex set: if $\mathbf{u}$ and $\mathbf{w}$ satisfy it, then for $0 \le t \le 1$,

$$\mathbf{a}^\top \left(t\mathbf{u} + (1-t)\,\mathbf{w}\right) = t\,\mathbf{a}^\top \mathbf{u} + (1-t)\,\mathbf{a}^\top \mathbf{w} \le c \tag{2.2.11.1}$$

The feasible region is the intersection of the half-planes given by the constraints, and an intersection of convex sets is convex. So the feasible region is convex.

Step 2: graph and corner points. For a linear objective over a convex feasible region, the optimum (maximum or minimum) lies at a corner point of the region. Plot the boundary lines $3x + y = 6$, $x = 1$, $x = 0$ and $y = 0$ (see Fig. 2.2.11.1). The corner points are the intersections of pairs of boundary lines that satisfy all the constraints,

$$(0,0)\,,\ (1,0)\,,\ (1,3)\,,\ (0,6) \tag{2.2.11.2}$$

Corner point (x, y)	$Z = 5x + 4y$
$(0, 0)$	0
$(1, 0)$	5
$(1, 3)$	17
$(0, 6)$	24

TABLE 2.2.11: Value of Z at the corner points

Step 3: optimise with `cvxpy`. Since the problem is convex, `cvxpy` finds the optimum for us. With the variables `x`, `y`, the constraints `[3*x + y <= 6, x <= 1, x >= 0, y >= 0]` and

```
objective = cp.Maximize(5*x + 4*y)
problem = cp.Problem(objective, constraints)
problem.solve()
```

the solver returns the same optimum as the corner points,

$$Z_{\max} = 24 \text{ at } (x, y) = (0, 6) \tag{2.2.11.3}$$

The following code checks the convexity, finds the corner points and solves the problem with `cvxpy`

12/codes/24.py

The following code generates Fig. 2.2.11.1.

12/codes/24plot.py

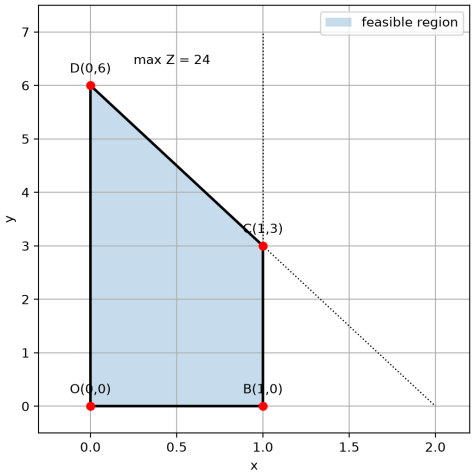


Fig. 2.2.11.1: Feasible region and corner points

2.2.12 A box contains 10 blocks: two marked 0, three marked 1, four marked 2 and one marked 3. A block is drawn at random. Find the probability distribution of the number X on the block and its mean. (12, 2025)

Solution: Each block is equally likely, so

$$\Pr(X = x) = \frac{\text{number of blocks marked } x}{10} \tag{2.2.12.1}$$

and the pmf is

x	0	1	2	3
$\Pr(X = x)$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{1}{10}$

TABLE 2.2.12: Probability mass function of X

The probabilities sum to 1. The mean of X is

$$E[X] = \sum_x x \Pr(X = x) = 0\left(\frac{2}{10}\right) + 1\left(\frac{3}{10}\right) + 2\left(\frac{4}{10}\right) + 3\left(\frac{1}{10}\right) \tag{2.2.12.2}$$

$$= \frac{14}{10} = 1.4 \tag{2.2.12.3}$$

The following program simulates 10^6 draws of a block and compares the relative frequencies and the mean with the pmf

12/codes/25.c

The following code makes a stem plot of the simulated relative frequencies and the pmf, see Fig. 2.2.12.1.

12/codes/25plot.py

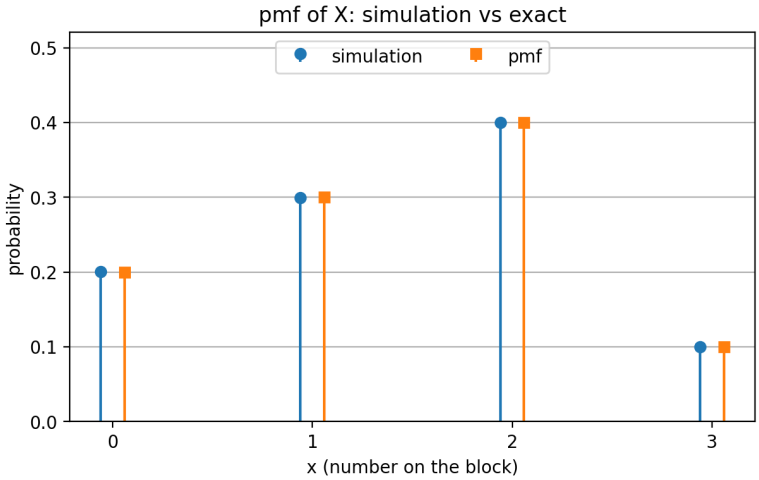


Fig. 2.2.12.1: Stem plot of the simulated relative frequencies and the pmf of X

2.2.13 Two ants move along the lines $2x + 5y - 1 = 0$ and $3x + 2y - 7 = 0$. Using the matrix method, find the point where their paths meet. (12, 2025)

Solution: In matrix form $\mathbf{Ax} = \mathbf{b}$ with the augmented matrix $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$, Gaussian elimination gives

$$\begin{pmatrix}2 & 5 & 1\\ 3 & 2 & 7\end{pmatrix} \xleftrightarrow{R_1 \leftarrow R_1/2} \begin{pmatrix}1 & \frac{5}{2} & \frac{1}{2}\\ 3 & 2 & 7\end{pmatrix} \tag{2.2.13.1}$$

$$\xleftrightarrow{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix}1 & \frac{5}{2} & \frac{1}{2}\\ 0 & -\frac{11}{2} & \frac{11}{2}\end{pmatrix} \tag{2.2.13.2}$$

$$\xleftrightarrow{R_2 \leftarrow -\frac{2}{11}R_2} \begin{pmatrix}1 & \frac{5}{2} & \frac{1}{2}\\ 0 & 1 & -1\end{pmatrix} \tag{2.2.13.3}$$

$$\xleftrightarrow{R_1 \leftarrow R_1 - \frac{5}{2}R_2} \begin{pmatrix}1 & 0 & 3\\ 0 & 1 & -1\end{pmatrix} \tag{2.2.13.4}$$

Both $\mathbf{A}$ and $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$ have rank 2, equal to the number of unknowns, so the lines meet at exactly one point,

$$\mathbf{x} = \begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}3\\ -1\end{pmatrix} \tag{2.2.13.5}$$

Check: $2\,(3) + 5\,(-1) = 1$ and $3\,(3) + 2\,(-1) = 7$. The paths meet at $(3, -1)$. The following code verifies the solution

12/codes/27.py

The following code generates Fig. 2.2.13.1.

12/codes/27plot.py

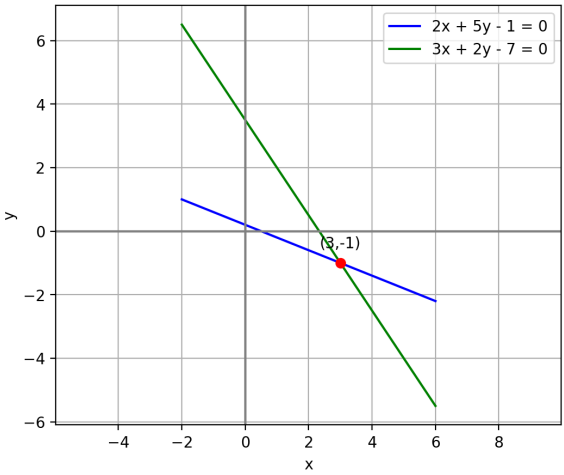


Fig. 2.2.13.1: The two paths and their common point $(3, -1)$

2.2.14 Minimise $Z = 18x + 10y$ subject to $4x + y \geq 20$, $2x + 3y \geq 30$, $x \geq 0$, $y \geq 0$. (12, 2025)

Solution: Step 1: the feasible region is convex. Each constraint is a half-plane $\mathbf{a}^\top \mathbf{x} \leq c$, and a half-plane is a convex set: if $\mathbf{u}$ and $\mathbf{w}$ satisfy it, then for $0 \leq t \leq 1$,

$$\mathbf{a}^\top (t\mathbf{u} + (1-t)\,\mathbf{w}) = t\,\mathbf{a}^\top \mathbf{u} + (1-t)\,\mathbf{a}^\top \mathbf{w} \leq c \tag{2.2.14.1}$$

The feasible region is the intersection of the half-planes given by the constraints, and an intersection of convex sets is convex. So the feasible region is convex.

Step 2: graph and corner points. For a linear objective over a convex feasible region, the optimum (maximum or minimum) lies at a corner point of the region. Plot the boundary lines $4x + y = 20$, $2x + 3y = 30$, $x = 0$ and $y = 0$ (see Fig. 2.2.14.1). The lines $4x + y = 20$ and $2x + 3y = 30$ meet where $y = 20 - 4x$ and $2x + 3\,(20 - 4x) = 30$, i.e. at $(3, 8)$. The corner points of the unbounded feasible region are

$$(0, 20)\,,\ (3, 8)\,,\ (15, 0) \tag{2.2.14.2}$$

Corner point (x, y)	$Z = 18x + 10y$
$(0, 20)$	200
$(3, 8)$	134
$(15, 0)$	270

TABLE 2.2.14: Value of Z at the corner points

Because the region is unbounded, the smallest corner value 134 is the minimum only if the half-plane $18x + 10y < 134$ has no point in common with the feasible region, which `cvxpy` confirms below.

$$Z_{\min} = 134 \text{ at } (x, y) = (3, 8) \tag{2.2.14.3}$$

The following code checks the convexity, finds the corner points and solves the problem with `cvxpy`

12/codes/30.py

The following code generates Fig. 2.2.14.1.

12/codes/30plot.py

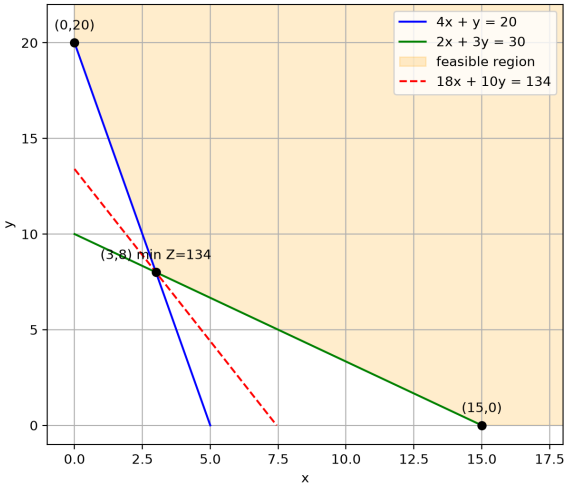


Fig. 2.2.14.1: Unbounded feasible region and the line $Z = 134$

2.2.15 Let $\mathbf{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\mathbf{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\mathbf{c} = \lambda\hat{i} - 2\hat{j} - 3\hat{k}$. Find λ if the scalar product of $\mathbf{a}$ with the unit vector along $\mathbf{b} + \mathbf{c}$ is 1. (12, 2025)

Solution:

$$\mathbf{b}+\mathbf{c}=\begin{pmatrix}\lambda+2\\-6\\2\end{pmatrix},\qquad |\mathbf{b}+\mathbf{c}|=\sqrt{(\lambda+2)^2+40} \tag{2.2.15.1}$$

The scalar product with $\mathbf{a}=\begin{pmatrix}1\\-1\\2\end{pmatrix}$ is

$$\frac{\mathbf{a}^\top(\mathbf{b}+\mathbf{c})}{|\mathbf{b}+\mathbf{c}|}=\frac{\lambda+2+6+4}{\sqrt{(\lambda+2)^2+40}}=\frac{\lambda+12}{\sqrt{(\lambda+2)^2+40}}=1 \tag{2.2.15.2}$$

Squaring,

$$(\lambda+12)^2=(\lambda+2)^2+40 \implies 20\lambda=-100 \implies \lambda=-5 \tag{2.2.15.3}$$

Check: $\lambda+12=7>0$ and $\sqrt{9+40}=7$, so the unsquared equation holds. The following code verifies the solution

12/codes/31A.py

2.2.16 Find the shortest distance between the lines $\mathbf{r}=\left(2\hat{i}-\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-2\hat{j}+3\hat{k}\right)$ and $\mathbf{r}=\left(\hat{i}+4\hat{k}\right)+\mu\left(3\hat{i}-6\hat{j}+9\hat{k}\right)$. (12, 2025)

Solution: The lines are

$$\mathbf{x}=\mathbf{A}+\kappa_1\mathbf{m}_1,\quad \mathbf{x}=\mathbf{B}+\kappa_2\mathbf{m}_2 \tag{2.2.16.1}$$

$$\mathbf{A}=\begin{pmatrix}2\\-1\\3\end{pmatrix},\ \mathbf{B}=\begin{pmatrix}1\\0\\4\end{pmatrix},\ \mathbf{m}_1=\begin{pmatrix}1\\-2\\3\end{pmatrix},\ \mathbf{m}_2=\begin{pmatrix}3\\-6\\9\end{pmatrix} \tag{2.2.16.2}$$

With $\mathbf{M}=\begin{pmatrix}\mathbf{m}_1 & \mathbf{m}_2\end{pmatrix}$, row reduction gives

$$\begin{pmatrix}1&3\\-2&-6\\3&9\end{pmatrix}\xleftrightarrow[R_3\leftarrow R_3-3R_1]{R_2\leftarrow R_2+2R_1}\begin{pmatrix}1&3\\0&0\\0&0\end{pmatrix} \tag{2.2.16.3}$$

so $\operatorname{rank}(\mathbf{M})=1$ and the lines are parallel, with $\mathbf{m}_2=3\mathbf{m}_1$. The squared distance between a point of each line is

$$\|\mathbf{A}+\kappa_1\mathbf{m}_1-\mathbf{B}-\kappa_2\mathbf{m}_2\|^2=\|(\mathbf{A}-\mathbf{B})+\kappa\mathbf{m}_1\|^2,\quad \kappa=\kappa_1-3\kappa_2 \tag{2.2.16.4}$$

which is least when its derivative with respect to κ vanishes, i.e. for the normal equation

$$\mathbf{m}_1^\top\mathbf{m}_1\,\kappa=\mathbf{m}_1^\top(\mathbf{B}-\mathbf{A}) \implies 14\kappa=\begin{pmatrix}1&-2&3\end{pmatrix}\begin{pmatrix}-1\\1\\1\end{pmatrix}=0 \implies \kappa=0 \tag{2.2.16.5}$$

Hence the closest point on the first line is $\mathbf{A}$ itself and the shortest distance is

$$d = \|\mathbf{A} - \mathbf{B}\| = \left\| \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \right\| = \sqrt{3} \tag{2.2.16.6}$$

The following code verifies the solution, including the least squares solution used in matgeo

12/codes/31B.py

2.2.17 Evaluate $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx$. (12, 2025)

Solution: Since $\frac{\tan x}{\sec x + \tan x} = \frac{\sin x}{1+\sin x}$, the integral is $I = \int_0^{\pi} \frac{x \sin x}{1+\sin x}\, dx$. Using $\int_0^{\pi} x g(x)\, dx = \frac{\pi}{2} \int_0^{\pi} g(x)\, dx$ when $g(\pi - x) = g(x)$,

$$I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1+\sin x}\, dx = \frac{\pi}{2} \int_0^{\pi} \left(1 - \frac{1}{1+\sin x}\right) dx \tag{2.2.17.1}$$

Rationalising, $\frac{1}{1+\sin x} = \sec^2 x - \sec x \tan x$, whose antiderivative is $\tan x - \sec x$. Over $[0, \pi]$ this gives $\int_0^{\pi} \frac{dx}{1+\sin x} = 2$ (the point $x = \frac{\pi}{2}$ is handled by taking limits on either side). Hence

$$I = \frac{\pi}{2}(\pi - 2) \approx 1.7932 \tag{2.2.17.2}$$

To evaluate the integral numerically, divide $[0, \pi]$ into N strips of width $h = \frac{\pi}{N}$ with nodes $x_n = nh$ and let $g(x) = \frac{x \sin x}{1+\sin x}$. Define the running integral

$$I[n] = \int_0^{x_n} g(x)\, dx, \quad I[0] = 0 \tag{2.2.17.3}$$

The strip between x_n and x_{n+1} is approximated by a trapezoid with parallel sides $g(x_n)$ and $g(x_{n+1})$ and width h, whose area is $\frac{h}{2}(g(x_n) + g(x_{n+1}))$. Hence I satisfies the difference equation

$$I[n+1] = I[n] + \frac{h}{2}(g(x_n) + g(x_{n+1})), \quad I[0] = 0 \tag{2.2.17.4}$$

and the integral is $I[N]$. Summing the equation over $n = 0, \ldots, N-1$ gives the trapezoidal rule

$$I[N] = \frac{h}{2}\left(g(x_0) + 2\sum_{n=1}^{N-1} g(x_n) + g(x_N)\right) \tag{2.2.17.5}$$

Iterating the difference equation gives $I[12] = 1.7754$, $I[100] = 1.79295$ and $I[1000] = 1.79321$, approaching 1.79321. The following code verifies the solution

12/codes/32B.py

The following code generates Fig. 2.2.17.1.

12/codes/32Bplot.py

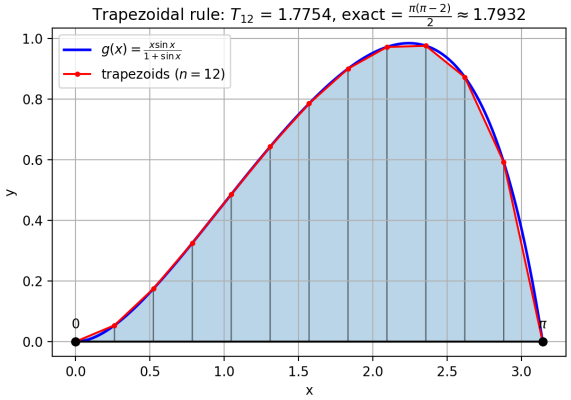


Fig. 2.2.17.1: Trapezoidal rule ($n = 12$) for $\frac{x \sin x}{1+\sin x}$ on $[0, \pi]$

2.2.18 Solve $\dfrac{dy}{dx} = \cos x - 2y$ (12, 2025)

Solution: Rewrite as $y' + 2y = \cos x$ and let $Y(s) = \mathcal{L}\{y\}$, $y(0) = y_0$. Since $\mathcal{L}\{y'\} = sY - y_0$ and $\mathcal{L}\{\cos x\} = \frac{s}{s^2+1}$,

$$(s+2)\,Y(s) = y_0 + \frac{s}{s^2+1} \implies Y(s) = \frac{y_0}{s+2} + \frac{s}{(s+2)(s^2+1)} \tag{2.2.18.1}$$

Partial fractions: $\frac{s}{(s+2)(s^2+1)} = \frac{-\frac{2}{5}}{s+2} + \frac{\frac{2}{5}s+\frac{1}{5}}{s^2+1}$. Using $\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2x}$, $\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos x$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin x$,

$$y(x) = \left(y_0 - \frac{2}{5}\right)e^{-2x} + \frac{2}{5}\cos x + \frac{1}{5}\sin x = Ce^{-2x} + \frac{2\cos x + \sin x}{5} \tag{2.2.18.2}$$

No initial condition is given, so the constant C is arbitrary. The following code verifies the solution

12/codes/34.py

2.2.19 Find the points on the line $\dfrac{2x+4}{6} = \dfrac{y+1}{2} = \dfrac{-2z+6}{-4}$ at a distance of $3\sqrt{2}$ units from the point $P(1,2,3)$. (12, 2025)

Solution: Let the common ratio be κ. Then $x = 3\kappa - 2$, $y = 2\kappa - 1$, $z = 2\kappa + 3$, which is the parametric form

$$\mathbf{x} = \mathbf{A} + \kappa\mathbf{m}, \quad \mathbf{A} = \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} \tag{2.2.19.1}$$

A point of the line is at distance $3\sqrt{2}$ from $\mathbf{P} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ when $\|\mathbf{A} + \kappa\mathbf{m} - \mathbf{P}\|^2 = 18$.

With $\mathbf{d} = \mathbf{A} - \mathbf{P} = \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix}$,

$$\left(\mathbf{m}^\top \mathbf{m}\right)\kappa^2 + 2\left(\mathbf{m}^\top \mathbf{d}\right)\kappa + \mathbf{d}^\top \mathbf{d} = 18 \implies 17\kappa^2 - 30\kappa + 18 = 18 \tag{2.2.19.2}$$

$$\kappa\left(17\kappa - 30\right) = 0 \implies \kappa = 0 \text{ or } \kappa = \frac{30}{17} \tag{2.2.19.3}$$

The required points are

$$\mathbf{Q}_1 = \mathbf{A} = \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}, \qquad \mathbf{Q}_2 = \mathbf{A} + \frac{30}{17}\mathbf{m} = \begin{pmatrix} \frac{56}{17} \\ \frac{43}{17} \\ \frac{111}{17} \end{pmatrix} \tag{2.2.19.4}$$

The following code verifies the solution

12/codes/35A.py

The following code generates Fig. 2.2.19.1.

12/codes/35Aplot.py

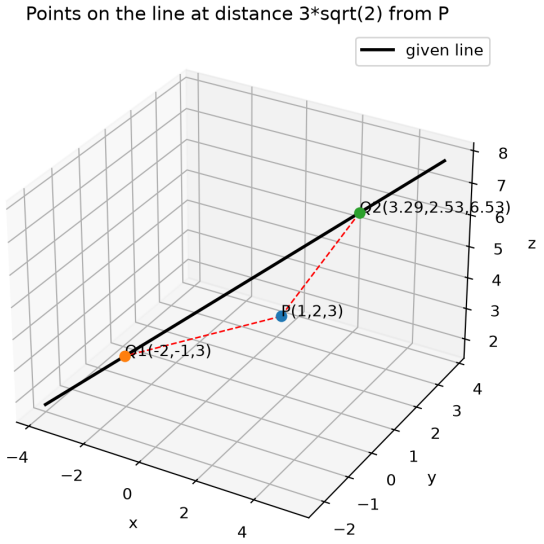


Fig. 2.2.19.1: Points Q_1, Q_2 on the line at distance $3\sqrt{2}$ from P

2.2.20 Find the image of the point $P\left(-1,5,2\right)$ in the line $\frac{2x-4}{2}=\frac{y}{2}=\frac{2-z}{3}$, and the length PP'. (12, 2025)

Solution: Let the common ratio be κ. Then $x=\kappa+2$, $y=2\kappa$, $z=2-3\kappa$, which is the parametric form

$$\mathbf{x}=\mathbf{A}+\kappa\mathbf{m},\quad \mathbf{A}=\begin{pmatrix}2\\0\\2\end{pmatrix},\quad \mathbf{m}=\begin{pmatrix}1\\2\\-3\end{pmatrix} \tag{2.2.20.1}$$

The foot of the perpendicular $\mathbf{H}=\mathbf{A}+\kappa_0\mathbf{m}$ from $\mathbf{P}=\begin{pmatrix}-1\\5\\2\end{pmatrix}$ satisfies $\mathbf{m}^\top\left(\mathbf{H}-\mathbf{P}\right)=0$, so

$$\kappa_0=\frac{\mathbf{m}^\top\left(\mathbf{P}-\mathbf{A}\right)}{\mathbf{m}^\top\mathbf{m}}=\frac{\begin{pmatrix}1&2&-3\end{pmatrix}\begin{pmatrix}-3\\5\\0\end{pmatrix}}{14}=\frac{7}{14}=\frac{1}{2}\implies\mathbf{H}=\begin{pmatrix}\frac{5}{2}\\1\\\frac{1}{2}\end{pmatrix} \tag{2.2.20.2}$$

Since $\mathbf{H}$ is the mid-point of $\mathbf{P}$ and its image $\mathbf{P}'$,

$$\mathbf{P}'=2\mathbf{H}-\mathbf{P}=\begin{pmatrix}5\\2\\1\end{pmatrix}-\begin{pmatrix}-1\\5\\2\end{pmatrix}=\begin{pmatrix}6\\-3\\-1\end{pmatrix} \tag{2.2.20.3}$$

$$PP'=\left\|\mathbf{P}'-\mathbf{P}\right\|=\left\|\begin{pmatrix}7\\-8\\-3\end{pmatrix}\right\|=\sqrt{49+64+9}=\sqrt{122} \tag{2.2.20.4}$$

The following code verifies the solution

12/codes/35B.py

The following code generates Fig. 2.2.20.1.

12/codes/35Bplot.py

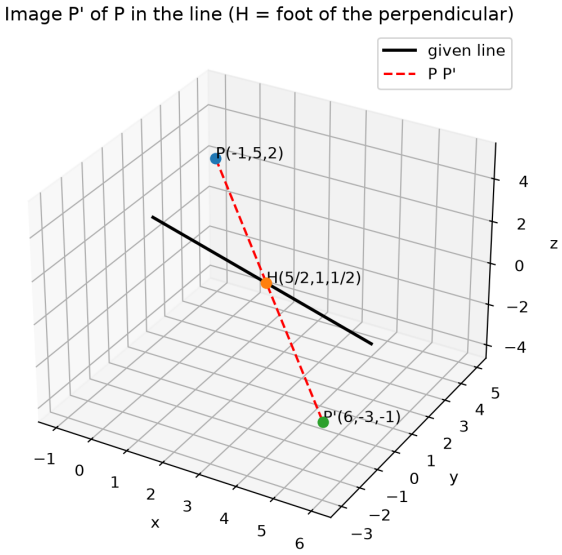


Fig. 2.2.20.1: Point P, its image P' and the foot H

2.2.21 **Case study (health survey).** Of 1000 people, 700 have good health (A_1), 200 average health (A_2) and 100 poor health (A_3). The probabilities of contracting the disease D are $\Pr(D \mid A_1) = 0.25$, $\Pr(D \mid A_2) = 0.35$ and $\Pr(D \mid A_3) = 0.50$. (i) Find the probability that a randomly selected person has contracted the disease. (ii) Find the probability that a person who has not contracted the disease belongs to category A_1. (12, 2025)

Solution: Let $X = 1, 2, 3$ denote the category A_X. The pmf of X and the conditional probabilities of the disease are

x	1	2	3
$p_X(x) = \Pr(A_x)$	0.70	0.20	0.10
$\Pr(D \mid A_x)$	0.25	0.35	0.50
$\Pr(D' \mid A_x) = 1 - \Pr(D \mid A_x)$	0.75	0.65	0.50

TABLE 2.2.21: pmf of the category and conditional probabilities

The joint pmf follows from $\Pr(A_x \cap D) = p_X(x) \Pr(D \mid A_x)$:

x	1	2	3
$\Pr(A_x \cap D)$	0.175	0.070	0.050
$\Pr(A_x \cap D')$	0.525	0.130	0.050

TABLE 2.2.21: Joint pmf of the category and the disease

(i) Summing over the categories (law of total probability),

$$\Pr(D) = 0.175 + 0.070 + 0.050 = 0.295 \tag{2.2.21.1}$$

(ii) Similarly $\Pr(D') = 0.525 + 0.130 + 0.050 = 0.705 = 1 - 0.295$. By Bayes' theorem,

$$\Pr(A_1 \mid D') = \frac{\Pr(A_1 \cap D')}{\Pr(D')} = \frac{0.525}{0.705} = \frac{35}{47} \approx 0.7447 \tag{2.2.21.2}$$

The following program simulates the category and the disease of 10^6 people and compares the relative frequencies with these values

12/codes/38.c

The following code makes a stem plot of the simulated relative frequencies and the pmf, see Fig. 2.2.21.1.

12/codes/38plot.py

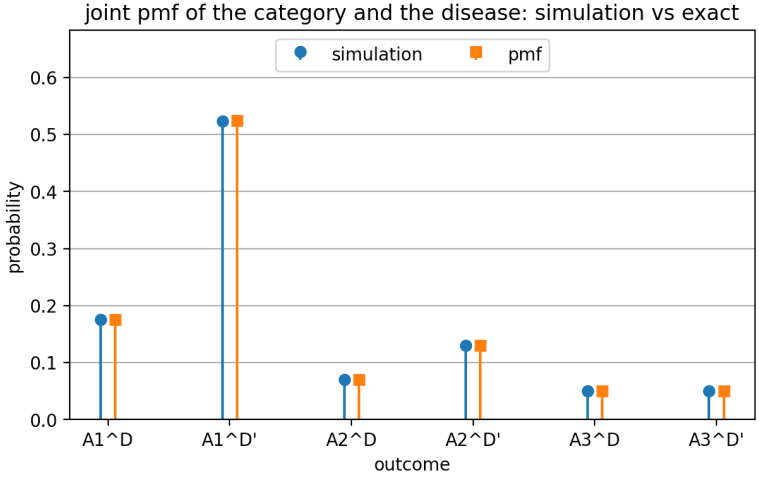


Fig. 2.2.21.1: Stem plot of the simulated relative frequencies and the joint pmf of the category and the disease